

PHY 615-2012

PROBLEM SET #6

Problem 1:

$$V(x) = \frac{\lambda}{4} \left(x^4 + ax^3 - \frac{1}{2}a^2x^2 \right)$$

$$V'(x) = \frac{\lambda}{4} (4x^3 + 3ax^2 - a^2x) = 0$$

$$V''(x) = \frac{\lambda}{4} (12x^2 + 6ax - a^2) = 0$$

Extrema:

$$x=0 \quad 4x^2 + 3ax - a^2 = 0$$

$$(4x-a)(x+a) = 0$$

$$\text{ie } x = 0, a/4, -a.$$

$$V''(0) = -\frac{\lambda a^2}{4} < 0 \rightarrow \text{maxima}$$

$$V''(-a) = \frac{\lambda}{4} (12a^2 - 6a^2 - a^2) = \frac{5\lambda a^2}{4} > 0$$

$$V''(a/4) = \frac{\lambda}{4} \frac{5a^2}{4} > 0$$

$$V(-a) = -\frac{\lambda a^4}{8}$$

$$V(a/4) = -\frac{3\lambda a^4}{1024}$$

$$\text{Thus } V(-a) < V(a/4).$$

$\Rightarrow$ Global minimum is $x = -a$

The trial wavefunction is

$$\psi(x) = \left(\frac{\beta}{\pi}\right)^{1/4} \exp \left[-\beta(x+a)^2/2 \right]$$

$$\left\langle \frac{p^2}{2m} \right\rangle = -\frac{1}{2m} \left(\frac{\beta}{\pi} \right)^{1/2} \int_{-\infty}^{\infty} dx e^{-\beta(x+a)^2/2} \frac{d^2}{dx^2} e^{-\beta(x+a)^2/2}$$

In integral shift $x \rightarrow x-a$

$$\begin{aligned} \left\langle \frac{p^2}{2m} \right\rangle &= -\frac{1}{2m} \sqrt{\frac{\beta}{\pi}} \int_{-\infty}^{\infty} dx e^{-\beta x^2/2} \frac{d^2}{dx^2} e^{-\beta x^2/2} \\ &= \frac{1}{2m} \sqrt{\frac{\beta}{\pi}} \int_{-\infty}^{\infty} dx \left(\frac{d}{dx} e^{-\beta x^2/2} \right)^2 \\ &= \frac{1}{2m} \sqrt{\frac{\beta}{\pi}} \beta^2 \int_{-\infty}^{\infty} dx x^2 e^{-\beta x^2} \\ &= \frac{1}{2m} \sqrt{\frac{\beta}{\pi}} \beta^2 \frac{1}{\beta^{3/2}} \int_{-\infty}^{\infty} dy y^2 e^{-y^2} \\ &= \frac{\beta}{2m} \frac{1}{\sqrt{\pi}} \frac{\sqrt{\pi}}{2} = \frac{\beta}{4m} \end{aligned}$$

For the potential

$$\begin{aligned} \langle V \rangle &= \frac{\lambda}{4} \sqrt{\frac{\beta}{\pi}} \int_{-\infty}^{\infty} dx e^{-\beta(x+a)^2} \left[x^4 + ax^3 - \frac{1}{2}a^2x^2 \right] \\ &= \frac{\lambda}{4} \sqrt{\frac{\beta}{\pi}} \int_{-\infty}^{\infty} dx e^{-\beta x^2} \left[(x-a)^4 + a(x-a)^3 - \frac{1}{2}a^2(x-a)^2 \right] \\ &= \frac{\lambda}{4} \sqrt{\frac{\beta}{\pi}} \sqrt{\frac{\pi}{\beta}} \left[\frac{3}{4\beta^2} + \frac{5a^2}{4\beta} - \frac{1}{2}a^4 \right] \end{aligned}$$

Thus

$$\begin{aligned}\langle E \rangle &= \frac{\beta}{4m} + \frac{\lambda}{4} \left(\frac{3}{4\beta^2} + \frac{5a^2}{4\beta} - \frac{1}{2}a^4 \right) \\ &= \frac{\beta}{4m} + \frac{\lambda}{16\beta^2} (3 + 5a^2\beta - 2\beta^2a^4)\end{aligned}$$

Thus a natural variable is $\beta a^2 = y$

$$\langle E \rangle = \frac{1}{4ma^2} \left[y + \frac{\lambda a^6}{4y^2} (3 + 5y - 2y^2) \right]$$

Thus eqn determining y is

Define $\alpha = \lambda ma^6$

$$1 - \frac{6\alpha}{4y^3} - \frac{5\alpha}{4y^2} = 0$$

$$4y^3 - 6\alpha - 5\alpha y = 0$$

$$\text{For } \alpha = 1 \quad (2y-3)(2y^2+3y+2) = 0$$

Solutions of $2x^2+3x+2=0$ are complex. ($3^2 - 4 \cdot 4 < 0$)

$\Rightarrow$ only real solution is $y = 3/2$

For this y

$$E_0 = \frac{1}{2ma^2} \left(\frac{13}{12} \right)$$

Problem 2: (16.2.1)

$$W(E) = \int_{x'}^x dx'' \sqrt{2m(E - V(x''))}$$

$$\frac{\partial W(E)}{\partial E} = \int_{x'}^x dx'' \sqrt{\frac{m}{2}} \frac{1}{\sqrt{E - V(x'')}}$$

$$= \int_{x'}^x dx'' \frac{dt}{dx''} = \int_{t'}^t dt$$

$$= (t - t')$$

Problem 3 :

(i) Consider a trial wave function

$$\psi_\alpha(x) = \left(\frac{\alpha^2}{2\pi}\right)^{1/4} \exp\left[-\frac{\alpha^2 x^2}{4}\right]$$

Then

$$\langle \psi_\alpha | \frac{\hat{p}^2}{2m} | \psi_\alpha \rangle = \frac{\hbar^2 \alpha^2}{8m} \geq 0$$

Now consider

$$\langle \psi_\alpha | V(\hat{x}) | \psi_\alpha \rangle = -\frac{\alpha}{\sqrt{2\pi}} \int_{-\infty}^{\infty} V(x) e^{-\alpha^2 x^2/2} dx$$

Since $V(\hat{x})$ is negative everywhere.

If $\int_{-\infty}^{\infty} dx V(x)$ converges, $\langle \psi_\alpha | V | \psi_\alpha \rangle$ has a finite limit as $\alpha \rightarrow 0$.
In this case,

$$\langle E \rangle = \frac{\hbar^2 \alpha^2}{8m} - \alpha c$$

where c is some ⁺ve number.

$$\frac{\partial \langle E \rangle}{\partial \alpha} = 0 \Rightarrow \frac{\hbar^2 \alpha_0}{4m} = c \quad \alpha_0 = \frac{4mc}{\hbar^2}$$

$$\Rightarrow \langle E(\alpha_0) \rangle = -\frac{2mc^2}{\hbar^2} < 0.$$

Since $E_0 \leq \langle E \rangle(\alpha)$ (E_0 is 8nd state energy)
 This shows $E_0 < 0$. Thus there
 is at least one bound state.

~~(i)~~ If $\int_{-\infty}^{\infty} dx V(x)$ is divergent,
 clearly $\langle \psi_0 | H | \psi_0 \rangle < 0$, so that
 $E_0 < 0$ as well.

(ii) In 3d, consider once again
 the trial wave fn.

$$\psi_\alpha = \left(\frac{\alpha^2}{2\pi}\right)^{3/4} \exp\left[-\frac{\alpha^2 r^2}{4}\right]$$

Then

$$\langle \psi_\alpha | V(x) | \psi_\alpha \rangle \sim -\alpha^3$$

If the integral of V is now finite

$$\langle E \rangle = \frac{\hbar^2 \alpha^2}{8m} - \alpha^3 C$$

and now the value of $\langle E \rangle$ at
 extremum is zero or +ve

— so we cannot use this argument
 to show there is a bound state

In fact generically there may or
 may not be bound states.

Problem 4

$$(i) \quad \left\langle \frac{p^2}{2m} \right\rangle = - \frac{\alpha}{2m} \int_{-\infty}^{\infty} dx e^{-\alpha|x|} (e^{-\alpha|x|})''$$

$$(e^{-\alpha|x|})'' = (-\alpha(\theta(x) - \theta(-x))e^{-\alpha|x|})'$$

$$= (-2\alpha\delta(x) + \alpha^2)e^{-\alpha|x|}$$

$$\left(\begin{array}{l} \theta'(x) = \delta(x) \end{array} \right)$$

Thus

$$\left\langle \frac{p^2}{2m} \right\rangle = - \frac{\alpha}{2m} \left[-2\alpha + \alpha^2 \int_{-\infty}^{\infty} dx e^{-2\alpha|x|} \right]$$

$$= - \frac{\alpha}{2m} \left[-2\alpha + 2\alpha^2 \int_0^{\infty} dx e^{-2\alpha x} \right]$$

$$= - \frac{\alpha}{2m} (-2\alpha + \alpha)$$

$$= \alpha^2/2m$$

The potential energy is

$$V = qE[\theta(x) - \theta(-x)]x$$

$$= qE|x|$$

Thus

$$\langle V \rangle = qE \int_{-\infty}^{\infty} dx |x| e^{-2\alpha|x|} = \frac{qE}{2\alpha}$$

$\Rightarrow$

$$\langle E \rangle = \frac{\alpha^2}{2m} + \frac{qE}{2\alpha}$$

$$\frac{\partial \langle E \rangle}{\partial \alpha} = \frac{\alpha}{m} - \frac{qE}{2\alpha^2} \Rightarrow \alpha = \left(\frac{qEm}{2} \right)^{1/3}$$

$$E_0 = \frac{3}{2^{5/3}} \left[\frac{(qE)^2}{m} \right]^{1/3}$$

(ii) Bohr Sommerfeld:

Turning points are $x_0 = \pm \frac{e}{qE}$

where e is the energy

Thus

$$\int_{-e/qE}^{e/qE} dx \sqrt{2m(e - qEx)} = \left(n + \frac{1}{2}\right) \pi$$

$$\int_{-e/qE}^0 dx \sqrt{2m(e + qEx)} + \int_0^{e/qE} dx \sqrt{2m(e - qEx)} = \left(n + \frac{1}{2}\right) \pi$$

$$\frac{2}{3} \frac{e^{3/2} \sqrt{2m}}{qE} + \frac{2}{3} \frac{e^{3/2} \sqrt{2m}}{qE} = \left(n + \frac{1}{2}\right) \pi$$

Thus gives for $n=0$ (ground state)

$$e^{3/2} = \frac{3qE\pi}{8\sqrt{2m}} = \frac{3\pi}{8\sqrt{2}} \left(\frac{qE}{\sqrt{m}}\right)$$

$$e = \left(\frac{3\pi}{8\sqrt{2}}\right)^{2/3} \left(\frac{(qE)^2}{m}\right)^{1/3}$$

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Problem 45

For energy E the turning points are $\pm x_0$ where

$$E = V(x_0) = \frac{1}{2} m \omega^2 x_0^2$$

$$\Rightarrow x_0 = \sqrt{\frac{2E}{m\omega^2}}$$

(1) In the allowed region the WKB wave function is (for energy E)

$$\psi(x) = \frac{A}{\sqrt{p(x)}} \cos\left(\frac{1}{\hbar} \int_{-x_0}^x p(x') dx' - \frac{\pi}{4}\right)$$

$$\begin{aligned} \text{where } p(x) &= \sqrt{2m(E - \frac{1}{2}m\omega^2 x^2)} \\ &= \sqrt{2m \cdot \frac{1}{2}m\omega^2 (x_0^2 - x^2)} \\ &= m\omega \sqrt{x_0^2 - x^2} \end{aligned}$$

Quantization condition is

$$\int_{-x_0}^{x_0} dx' p(x') = (n + \frac{1}{2})\hbar$$

$$\Rightarrow m\omega x_0^2 \int_{-1}^1 dx \sqrt{1-x^2} = (n + \frac{1}{2})\hbar \pi$$

$$\frac{2E}{\omega} \cdot \frac{\pi}{2} = (n + \frac{1}{2})\hbar$$

$$\Rightarrow \boxed{E = (n + \frac{1}{2})\hbar\omega}$$

